

# RISHABH TRIPATHI

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## Summary

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- Computational biologist with an M.S. in Bioinformatics and hands-on experience turning large, messy sequencing data into results scientists can act on.
- At Camp4 Therapeutics, built and ran the production **NGS pipelines and QC tools** (nf-core Nextflow on AWS) behind the discovery team's daily RNA-seq and ChIP-seq work.
- Strong in **Python and R**, with reproducible, containerized workflows (Docker, Git CI/CD) and a from-scratch variant-calling pipeline validated to **99% accuracy** against a gold-standard reference.
- Experienced across **bulk, single-cell, and spatial RNA-seq**, variant calling and interpretation, statistical analysis, and interpretable machine learning.
- Collaborative teammate across wet-lab, chemistry, and data groups; picks up new methods quickly and genuinely enjoys building tools others rely on.

## Technical Skills

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**Programming:** Python (Pandas, NumPy, Scanpy, scikit-learn, TensorFlow), R (Bioconductor, DESeq2, edgeR, Shiny), SQL, Bash, Linux  
**NGS & Genomics:** Bulk & single-cell RNA-seq, ChIP-seq, Spatial Transcriptomics (Visium), Variant Calling & Interpretation, Differential Expression, Pathway Enrichment, ASO Screening Analysis  
**Pipelines & Infrastructure:** Nextflow, Docker, GitHub Actions (CI/CD), HPC & cloud environments, Reproducible workflow design  
**Cloud & Data:** AWS (EC2, S3, Batch), SQL/SQLite, Parquet, large-scale data processing  
**ML & Statistics:** Random Forest, Gradient Boosting, SVM, SHAP, statistical & differential analysis, dimensionality reduction, clustering  
**Visualization:** R Shiny, Streamlit, Tableau, Matplotlib, Seaborn  
**Molecular Modeling:** Molecular docking (AutoDock Vina, iGEMDOCK), homology modeling, Discovery Studio, SwissADME

## Education

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### Northeastern University Jan 2024 – May 2026

*Master of Science in Bioinformatics, GPA: 3.6, Boston, MA*

Relevant coursework: Statistics for Bioinformatics, Genomics in Bioinformatics, Data Collection/Storage/Retrieval, Omics, Transcriptomics.

### Loyola Academy Degree and PG College Aug 2020 – May 2023

*Bachelor of Science in Biotechnology, Chemistry, and Genetics, GPA: 3.1, Hyderabad, India*

**Certification:** Data Analytics Essentials, Cisco (Excel, SQL, Tableau), Credly verified

## Experience

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### Bioinformatics Co-op July 2025 – Dec 2025

*Camp4 Therapeutics Corporation Boston, MA*

#### NGS Data Processing & QC

- Ran production **nf-core Nextflow pipelines (AWS EC2)** for bulk RNA-seq and ChIP-seq supporting mouse time-course transcriptomics and gene-regulation studies, with **MultiQC** reporting.
- Built a **ChIP-seq data organization and QC tool** in R and Bash that scans processed files, auto-sorts by type, extracts quality metrics, and generates **genome-browser tracks** for the team.
- Designed and deployed **R Shiny apps** and automated QC reports so non-computational colleagues could explore NGS results directly.
- Added unit and regression tests and maintained reproducible test environments across Windows and Linux.

#### Drug Screening Analytics & Visualization

- Built and enhanced internal **R Shiny analytics tools** for ASO screening, adding automated parameter selection, target-gene filters, **EC50 dose-response** modeling, and filterable Bootstrap 5 dashboards.
- Developed an ASO chemistry visualization tool with sequence-notation toggles, coordinate-based alignment, and PDF export, built cross-functionally with chemistry and data teams.
- Designed interactive **dashboards** for real-time monitoring of compound history, experiment progress, and aggregated study metrics for the data team.

- Conducted an in-silico drug-discovery study to identify inhibitors of **TMPRSS4**, a serine protease overexpressed in multiple cancers, as a potential anti-cancer target.
- Built a **homology model** of TMPRSS4, prepared protein and ligand structures, and validated model quality with a **Ramachandran plot**.
- Screened a large phytochemical/natural-compound library (COCONUT, ZINC) via energy-based **molecular docking** (AutoDock Vina, iGEMDOCK), filtering by binding energy and predicted toxicity (SwissADME).
- Identified the **top 10 candidate inhibitors** with favorable binding and no predicted toxicity, and characterized ligand-receptor interactions (key residues, binding poses) in **Discovery Studio**.

## Selected Computational Biology Projects

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### Germline WGS Variant-Calling Workflow (GATK4 Best Practices)

May 2026 - Jul 2026

*Nextflow DSL2 • GATK4 • BWA-MEM • Docker • AWS Batch • hap.py • GIAB*

- Engineered a reproducible **FASTQ-to-VCF workflow** implementing **GATK4 Best Practices** with BWA-MEM, MarkDuplicates, BQSR, HaplotypeCaller GVCF joint genotyping, and hard filtering in modular **Nextflow DSL2**; containerized each process and configured **AWS Batch** execution.
- Benchmarked calls against the **GIAB HG002 truth set** with **hap.py**: **99.5% precision / 98.5% recall for SNPs (F1 0.99)** and **98.8% / 97.6% for INDELs (F1 0.98)** on chr20; Ti/Tv of about 2.0 further supported callset quality.
- Automated callset QC and reporting with **bcftools stats**, **MultiQC**, and a dependency-free **Python VCF-summary utility** for Ti/Tv, het/hom, and PASS-rate metrics; packaged a chr20 test profile for reproducible local and cloud runs.

### Automated RNA-Seq Quantification & QC Workflow

Feb 2026 - Apr 2026

*Nextflow DSL2 • FastQC • Trim Galore • Salmon • tximport • R • MultiQC*

- Automated paired-end RNA-seq processing in a **sample-sheet-driven Nextflow DSL2 workflow** spanning raw and post-trim FastQC, Trim Galore, Salmon indexing and quantification, and MultiQC reporting.
- Integrated **R/tximport** to consolidate per-sample Salmon outputs into transcript-count, TPM, and effective-length matrices ready for downstream differential-expression analysis.
- Structured reusable processes with **tuple-based channels and emitted outputs**, enabling consistent execution across nine *S. cerevisiae* samples representing three experimental conditions.

### Cell-Type-Resolved Spatial Transcriptomics Atlas (Mouse Brain, Visium HD)

Dec 2025 - Jan 2026

*Spatial transcriptomics • scRNA-seq integration • Deconvolution • Domain analysis*

- Integrated **Visium HD spatial RNA-seq** at 8  $\mu\text{m}$  and 16  $\mu\text{m}$  resolution with single-cell references to deconvolve cell types, transfer labels, and map high-resolution abundance patterns across tissue sections.
- Identified coherent spatial-expression domains using graph-based smoothing and **Leiden clustering**, then characterized domain-specific marker genes through nonparametric differential expression.
- Standardized QC, enrichment, and deliverables across UMI depth, gene complexity, mitochondrial content, and domain-level enrichment, exporting **AnnData, Parquet, CSV, dashboards, and per-sample QC HTML**.

### Single-Cell COVID-19 Immunoprofiling & Classification

Jan 2025 - Mar 2025

*Python • Scanpy • Harmony • pyDESeq2 • scikit-learn • TensorFlow • GSEAPy*

- Analyzed **approximately 37,000 PBMCs** from GSE149689 across 14 donor cohorts, applying Harmony/ComBat batch correction, Leiden clustering, and marker-based cell-type annotation in Scanpy.
- Resolved disease-associated shifts through **cell-type differential abundance and pseudobulk DESeq2**, recovering the severe-monocyte signature (S100A8/A9/A12 up; HLA class II down), lymphopenia, and dendritic-cell loss.
- Packaged preprocessing, integration, pseudobulk differential expression, and **GO/KEGG enrichment** into a modular CLI, with Random Forest, SVM, and TensorFlow models for PBMC classification.